Network Traffic Analysis Report

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Date: April 2026 | Tool: Wireshark 4.x | Interface: Wi-Fi

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1. Objective

To capture and analyze live home network traffic using Wireshark, identify the protocols in use, detect anomalous or suspicious patterns, and document findings in a structured format aligned with SOC analyst workflows.

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2. Tools & Environment

Field	Details
Tool Used	Wireshark 4.x (with Npcap 1.86 driver)
Operating System	Windows 11
Capture Interface	Wi-Fi (Home network)
Capture Duration	Approximately 10 minutes
Total Packets Captured	1,262 packets
Total Data Captured	181,987 bytes (~182 KB)
Capture File	capture1.pcap

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3. Methodology

The following steps were performed during this analysis:

- Installed Wireshark and Npcap on a Windows 11 machine
- Initiated a live packet capture on the active Wi-Fi interface
- Generated traffic by visiting multiple websites and running ping commands
- Applied display filters (dns, http, tcp.flags.syn==1) to isolate specific traffic types
- Analyzed Statistics > Protocol Hierarchy to understand traffic composition
- Reviewed Statistics > Conversations (TCP tab) to identify top communication endpoints
- Documented all findings with screenshots for evidence

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4. Findings

4.1 Protocol Distribution

The Protocol Hierarchy statistics revealed the following breakdown of captured traffic:

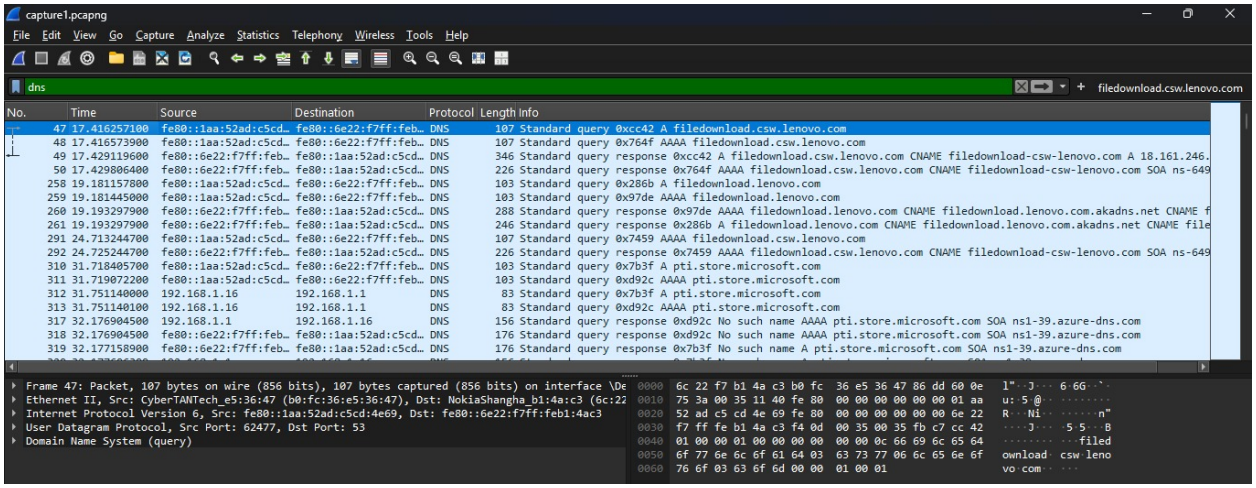
Protocol	Packets	% of Traffic
IPv6 / UDP / DNS	1,248	98.9%
IPv4 / UDP / DNS	14	1.1%

Observation: The overwhelming dominance of DNS traffic (98.9%) over IPv6 indicates the device was actively resolving domain names throughout the capture window, consistent with background services and browser activity.

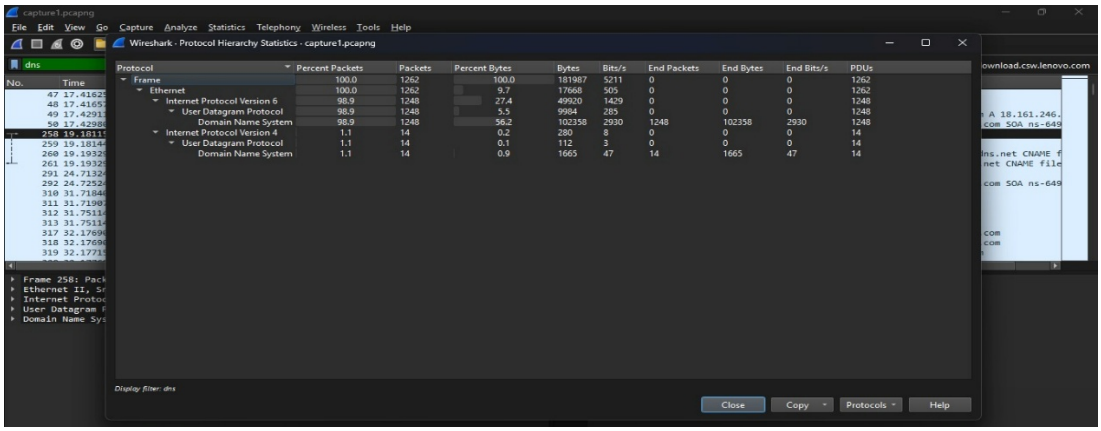
4.2 DNS Analysis

Applying the dns display filter revealed 1,262 DNS query and response packets. Key domains queried included:

- filedownload.csw.lenovo.com — Lenovo background software making download requests (AAAA + CNAME records observed)
- pti.store.microsoft.com — Microsoft Store service performing routine lookups
- Google NTP servers — Time synchronization traffic (GET /time/1/current)
- DNS rate observed: 2,930 bits/sec sustained throughout the capture



Protocol Hierarchy Statistics

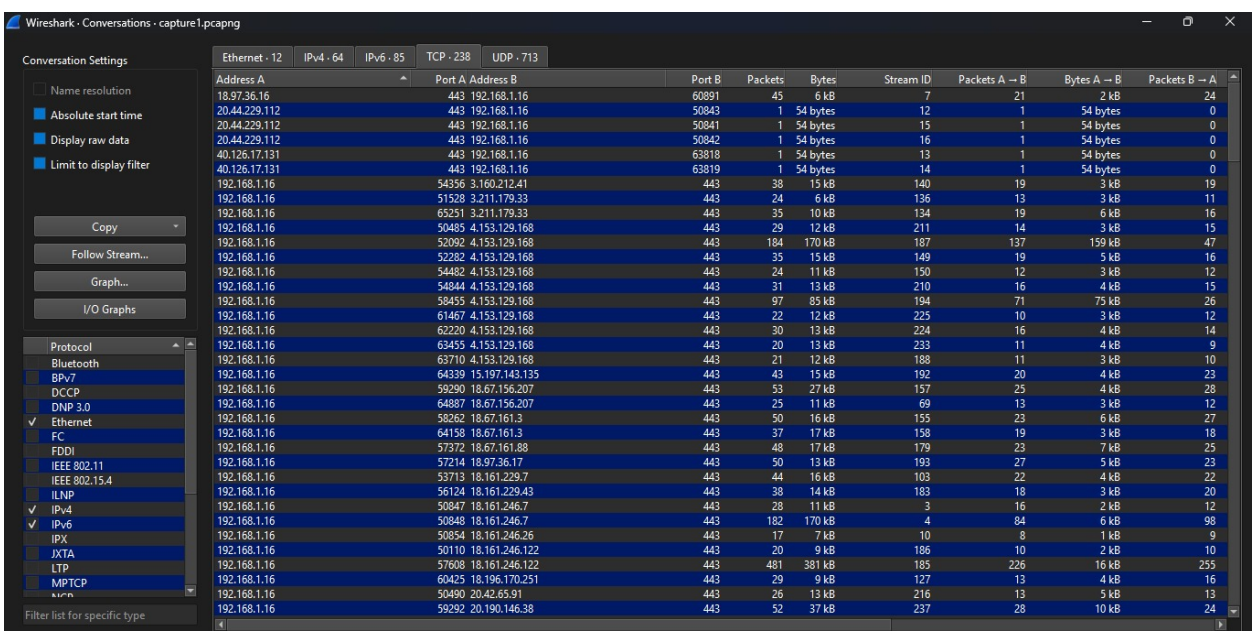


Security Finding: 3 DNS responses returned "No such name" (NXDOMAIN) for pti.store.microsoft.com — indicating a failed domain resolution. This is consistent with a misconfigured or deprecated Microsoft endpoint and does not indicate malicious activity but is worth flagging as a monitoring point.

4.3 TCP Conversations

The TCP Conversations tab (Statistics > Conversations) revealed 238 unique TCP streams. The top endpoints identified were:

- 18.161.246.122 — Akamai CDN: busiest endpoint with 481 packets and 381 KB transferred
- 4.153.129.168 — Microsoft Azure: multiple streams, up to 170 KB per stream
- 3.160.242.41 / 3.211.179.33 — Amazon AWS infrastructure
- 18.97.36.16 / 18.97.36.17 — Microsoft servers



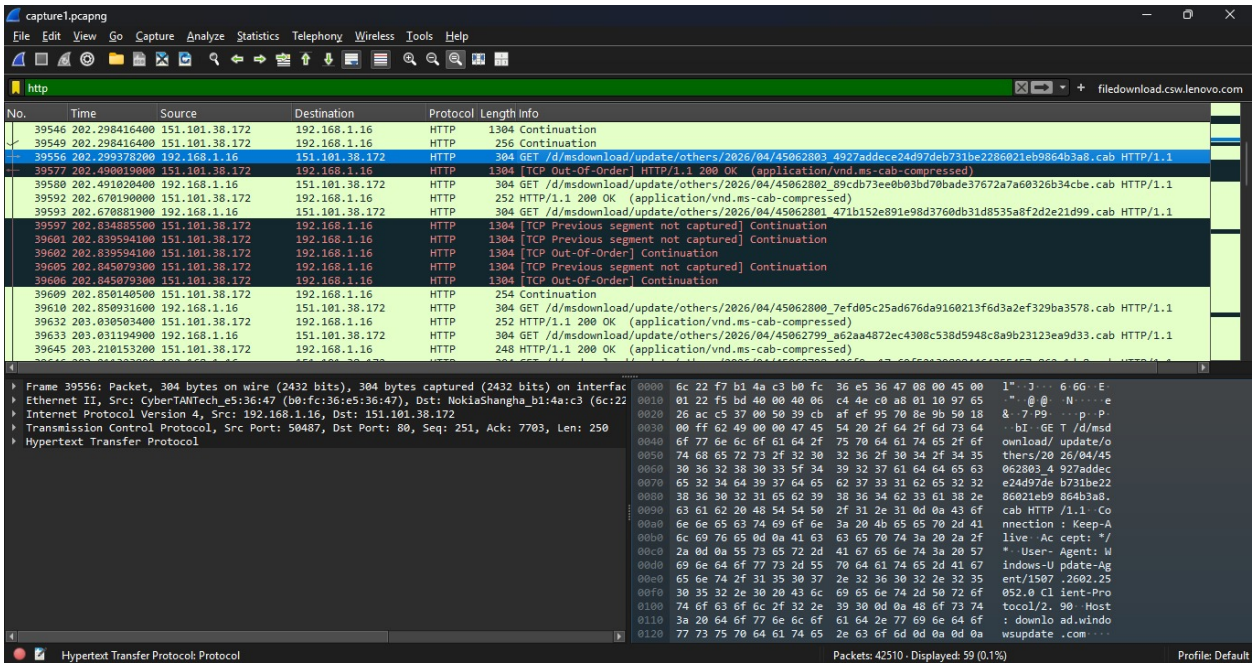
Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A
18.97.36.16	443	192.168.1.16	60891	45	6 kB	7	21	2 kB	24
20.44.229.112	443	192.168.1.16	50843	1	54 bytes	12	1	54 bytes	0
20.44.229.112	443	192.168.1.16	50841	1	54 bytes	15	1	54 bytes	0
20.44.229.112	443	192.168.1.16	50842	1	54 bytes	16	1	54 bytes	0
40.126.17.131	443	192.168.1.16	63818	1	54 bytes	13	1	54 bytes	0
40.126.17.131	443	192.168.1.16	63819	1	54 bytes	14	1	54 bytes	0
192.168.1.16	54356	3.160.212.41	443	38	15 kB	140	19	3 kB	19
192.168.1.16	51528	3.211.179.33	443	24	6 kB	136	13	3 kB	11
192.168.1.16	65251	3.211.179.33	443	35	10 kB	134	19	6 kB	16
192.168.1.16	50485	4.153.129.168	443	29	12 kB	211	14	3 kB	15
192.168.1.16	52092	4.153.129.168	443	184	170 kB	187	137	159 kB	47
192.168.1.16	52282	4.153.129.168	443	35	15 kB	149	19	5 kB	16
192.168.1.16	54482	4.153.129.168	443	24	11 kB	150	12	3 kB	12
192.168.1.16	54844	4.153.129.168	443	31	13 kB	210	16	4 kB	15
192.168.1.16	58455	4.153.129.168	443	97	85 kB	194	71	75 kB	26
192.168.1.16	61467	4.153.129.168	443	22	12 kB	225	10	3 kB	12
192.168.1.16	62220	4.153.129.168	443	30	13 kB	224	16	4 kB	14
192.168.1.16	63455	4.153.129.168	443	20	13 kB	233	11	4 kB	9
192.168.1.16	63710	4.153.129.168	443	21	12 kB	188	11	3 kB	10
192.168.1.16	64339	15.197.143.135	443	43	15 kB	192	20	4 kB	23
192.168.1.16	59290	18.67.156.207	443	53	27 kB	157	25	4 kB	28
192.168.1.16	64887	18.67.156.207	443	25	11 kB	69	13	3 kB	12
192.168.1.16	58262	18.67.161.3	443	50	16 kB	155	23	6 kB	27
192.168.1.16	64158	18.67.161.3	443	37	17 kB	158	19	3 kB	18
192.168.1.16	57372	18.67.161.88	443	48	17 kB	179	23	7 kB	25
192.168.1.16	57214	18.97.36.17	443	50	13 kB	193	27	5 kB	23
192.168.1.16	53713	18.161.229.7	443	44	16 kB	103	22	4 kB	22
192.168.1.16	56124	18.161.229.43	443	38	14 kB	183	18	3 kB	20
192.168.1.16	50847	18.161.246.7	443	28	11 kB	3	16	2 kB	12
192.168.1.16	50848	18.161.246.7	443	182	170 kB	4	84	6 kB	98
192.168.1.16	50854	18.161.246.26	443	17	7 kB	10	8	1 kB	9
192.168.1.16	50110	18.161.246.122	443	20	9 kB	186	10	2 kB	10
192.168.1.16	57608	18.161.246.122	443	481	381 kB	185	226	16 kB	255
192.168.1.16	60425	18.196.170.251	443	29	9 kB	127	13	4 kB	16
192.168.1.16	50490	20.42.65.91	443	26	13 kB	216	13	5 kB	13
192.168.1.16	59292	20.190.146.38	443	52	37 kB	237	28	10 kB	24

Security Finding: All 238 TCP conversations used Port 443 (HTTPS/TLS). No unencrypted TCP sessions were observed on sensitive ports. This indicates the device maintains good encryption hygiene for outbound connections.

4.4 HTTP Traffic

Applying the http filter revealed a small number of unencrypted HTTP packets, primarily consisting of:

- Microsoft Windows Update .cab file downloads from 151.101.38.172 (Fastly CDN)
- Google time synchronization request (GET /time/1/current) over HTTP/1.1
- TCP Out-of-Order and Previous Segment Not Captured events — normal behavior on a busy home network

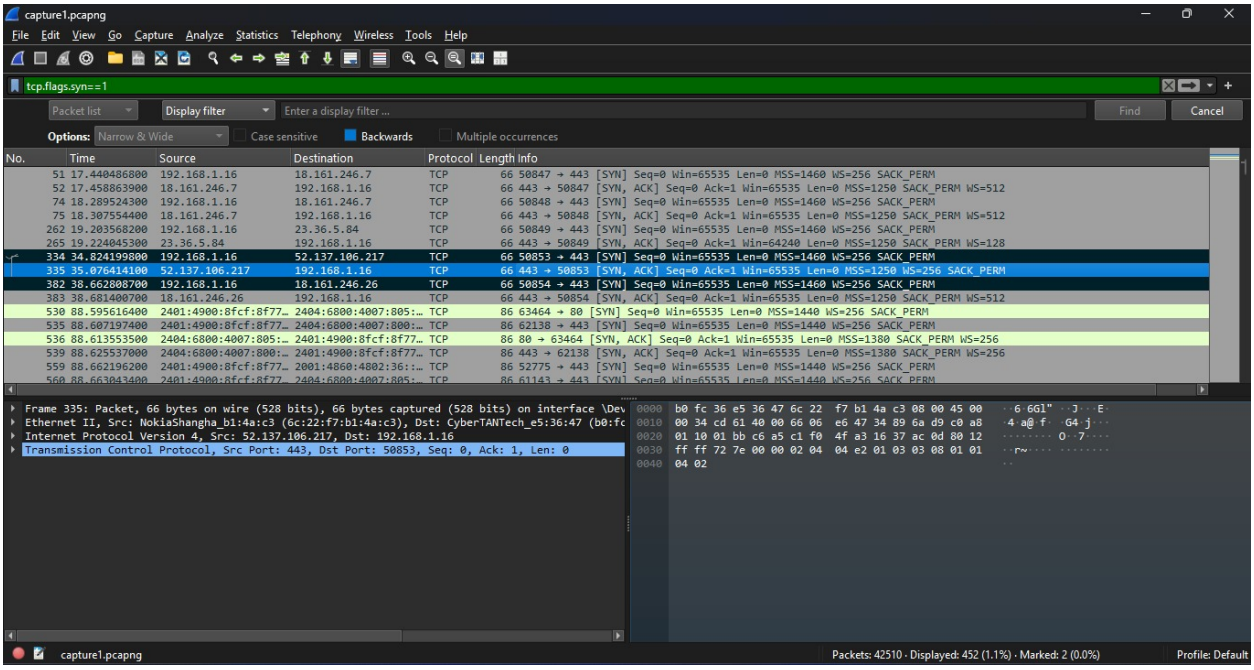


Observation: The presence of unencrypted HTTP for Windows Update delivery is a known Microsoft behavior for certain update metadata requests. No sensitive user data was observed in plaintext.

4.5 TCP SYN Analysis (New Connections)

Applying the filter tcp.flags.syn==1 revealed 15+ SYN packets representing new connection initiations. Key observations:

- All SYN packets received valid SYN-ACK responses — confirming successful three-way handshakes
- Destinations included Microsoft Azure (52.137.106.217) and Akamai CDN (18.161.246.26)
- No SYN flood pattern detected — no single destination received an abnormal volume of SYN packets
- No suspicious port scanning activity observed



5. Summary of Findings

Finding	Severity	Notes
NXDOMAIN DNS responses	Low	Deprecated Microsoft endpoint, no user data exposed
Unencrypted HTTP (Windows Update)	Low	Known Microsoft behavior for update metadata
All TCP on Port 443 (HTTPS)	Info (Good)	All 238 TCP streams used TLS encryption
No port scanning or SYN flood	Info (Good)	Clean SYN/SYN-ACK handshakes observed

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6. Conclusion

This network traffic analysis of a home Wi-Fi environment over approximately 10 minutes captured 1,262 packets totaling 182 KB of data. The traffic was predominantly DNS-based (98.9%), with all TCP connections using TLS encryption on port 443.

Two low-severity observations were noted: NXDOMAIN responses for a deprecated Microsoft endpoint, and unencrypted HTTP traffic for Windows Update metadata — both consistent with normal Windows operating system behavior. No indicators of compromise, port scanning, or suspicious outbound connections were detected.

This analysis demonstrates practical application of Wireshark for network traffic inspection, protocol analysis, and basic threat detection aligned with SOC analyst workflows.

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